

TECHNICAL EQUIPMENT DOCUMENTATION

Centrifugal Gas Compressor System

IDENTITY INFORMATION

Company_Name: Permian Basin Energy Solutions LLC

Field_Engineer: Marcus J. Whitfield

Field_Engineer_Id: PBE-FE-4827

Document_Version: 2.3.1

Machine_Name: Primary Gas Compression Unit - Platform A

Machine_Id: PBEC-COMP-2847-A

Installation_Date: 2021-03-15

Number_Of_Units_In_Machine: 3

UNIT 1 SPECIFICATIONS

Unit_Name: Centrifugal Compressor Stage 1 (Inlet)

Unit_Id: PBEC-COMP-2847-A-S1

Manufacture_Date: 2020-11-22

Manufacturer_Company_Name: Turbomachinery Dynamics International

Manufacturer_Company_Id: TDI-MFG-5634

Inventory_Quantity: 2 units in stock

Supplier_Information: Gulf Coast Industrial Supply, Houston TX | Contact: procurement@gcisupply.com | Lead Time: 8-10 weeks

DOMAIN SPECIFICATIONS:

- **Wellbore_Depth_Rating:** N/A (surface equipment)
- **Operating_Pressure_PSI:** 450-650 PSI inlet; 1,200-1,450 PSI discharge
- **Drilling_Fluid_Compatibility:** Dry gas only; moisture content <50 ppm
- **Known_Error_Codes:** E-401 (Surge condition), E-402 (Compressor stall), E-405 (Bearing temperature high)
- **Replacement_Part_Number:** TDI-CENT-S1-2847-REV-C
- **Maintenance_Schedule:** 500-hour inspections; 2,000-hour overhaul; annual vibration analysis

UNIT 2 SPECIFICATIONS

****Unit_Name:** Centrifugal Compressor Stage 2 (Intermediate)**

****Unit_Id:** PBEC-COMP-2847-A-S2**

****Manufacture_Date:** 2020-12-08**

****Manufacturer_Company_Name:** Turbomachinery Dynamics International**

****Manufacturer_Company_Id:** TDI-MFG-5634**

****Inventory_Quantity:** 1 unit in stock**

****Supplier_Information:** Gulf Coast Industrial Supply, Houston TX | Contact: procurement@gcisupply.com | Lead Time: 10-12 weeks**

****DOMAIN SPECIFICATIONS:****

- ****Wellbore_Depth_Rating:** N/A (surface equipment)**
- ****Operating_Pressure_PSI:** 1,200-1,450 PSI inlet; 2,800-3,200 PSI discharge**
- ****Drilling_Fluid_Compatibility:** Dry gas only; moisture content <50 ppm**
- ****Known_Error_Codes:** E-403 (Impeller erosion), E-404 (Seal degradation), E-406 (Discharge temperature exceedance)**
- ****Replacement_Part_Number:** TDI-CENT-S2-2847-REV-C**
- ****Maintenance_Schedule:** 500-hour inspections; 2,000-hour overhaul; quarterly seal integrity checks**

UNIT 3 SPECIFICATIONS

****Unit_Name:** Centrifugal Compressor Stage 3 (Discharge/Boost)**

****Unit_Id:** PBEC-COMP-2847-A-S3**

****Manufacture_Date:** 2021-01-14**

****Manufacturer_Company_Name:** Turbomachinery Dynamics International**

****Manufacturer_Company_Id:** TDI-MFG-5634**

****Inventory_Quantity:** 1 unit in stock**

****Supplier_Information:** Gulf Coast Industrial Supply, Houston TX | Contact: procurement@gcisupply.com | Lead Time: 12-14 weeks**

****DOMAIN SPECIFICATIONS:****

- ****Wellbore_Depth_Rating:** N/A (surface equipment)**
- ****Operating_Pressure_PSI:** 2,800-3,200 PSI inlet; 4,500-5,200 PSI discharge**
- ****Drilling_Fluid_Compatibility:** Dry gas only; moisture content <50 ppm**
- ****Known_Error_Codes:** E-407 (Thrust bearing wear), E-408 (Rotor imbalance), E-409 (Coupling misalignment)**
- ****Replacement_Part_Number:** TDI-CENT-S3-2847-REV-C**
- ****Maintenance_Schedule:** 500-hour inspections; 2,000-hour overhaul; semi-annual alignment verification**

SECTION 1 - USE CASE

- **Primary Function:** Three-stage centrifugal compressor system designed to compress natural gas from wellhead pressure (450 PSI) to pipeline injection pressure (5,200 PSI) for export and sales
- **Operational Role:** Intermediate processing equipment in gas production facility; operates continuously at 8,500-9,200 RPM with variable inlet guide vanes for flow control
- **Gas Throughput:** Designed capacity 85-95 MMSCFD (million standard cubic feet per day) with turndown capability to 60% minimum flow
- **Integration Point:** Receives inlet gas from separator train; discharges to dehydration unit and sales pipeline via check valve and relief systems
- **Redundancy Configuration:** Single primary unit with staged compression; backup compressor available for emergency operation per API RP 54 Section 8.2

SECTION 2 - FAILURE CAUSES

- **E-401 Surge Condition (Frequency: High):** Occurs when inlet flow drops below minimum stable flow threshold; caused by downstream blockage, anti-surge valve malfunction, or rapid load changes; results in reverse flow and compressor instability
- **E-405 Bearing Temperature Exceedance (Frequency: High):** Bearing degradation from inadequate lubrication, contaminated oil, or excessive radial/thrust loads; typical threshold 180°F; leads to catastrophic bearing failure if unaddressed within 4 hours
- **E-403 Impeller Erosion (Frequency: Medium):** Particulate contamination in inlet gas (sand, scale, corrosion products) causes blade erosion; reduces efficiency 2-5% per 500 hours; accelerated by moisture ingress and liquid carryover
- **E-408 Rotor Imbalance (Frequency: Medium):** Develops from blade erosion asymmetry, deposits on rotor surfaces, or manufacturing defects; vibration increases exponentially; detected via ISO 20816 monitoring at >7.1 mm/s velocity
- **E-409 Coupling Misalignment (Frequency: Low-Medium):** Thermal growth, foundation settling, or improper reassembly causes angular/parallel misalignment; induces excessive bearing loads and premature seal wear; detected via dial indicator checks during maintenance

SECTION 3 - SOLUTIONS

- **Surge Prevention & Control (API RP 54 8.3.2):** Install and maintain anti-surge recycle valve with minimum flow bypass; set recycle valve opening pressure 5% above minimum stable flow point; perform monthly functional tests; ensure inlet check valve operates freely; monitor inlet pressure continuously with redundant transmitters
- **Bearing Temperature Management (OSHA 1910.212):** Implement automated shutdown at 185°F bearing temperature; maintain oil analysis program with quarterly sampling (ISO 4406 cleanliness target 16/14/11); change bearing oil every 2,000 operating hours or per manufacturer specification; ensure cooling water flow >15 GPM at inlet temperature <85°F

- **Inlet Gas Filtration & Moisture Control:** Install 10-micron coalescing inlet filter with differential pressure indicator; replace filter elements when ΔP exceeds 5 PSI; maintain inlet gas dew point $< -20^{\circ}\text{F}$ via upstream dehydration; perform monthly visual inspection of filter housing for liquid accumulation
- **Vibration Monitoring & Alignment (API RP 54 8.4.1):** Conduct ISO 20816 vibration analysis every 500 operating hours; establish alarm thresholds at 6.5 mm/s (alert) and 7.5 mm/s (shutdown); perform laser alignment of motor-compressor coupling annually with tolerance $\pm 0.05''$ parallel and $\pm 0.02''$ angular; document all measurements in equipment logbook
- **Seal Integrity & Replacement (OSHA 1910.119 PSM):** Inspect mechanical seals quarterly for leakage (acceptable limit < 5 drops/minute); replace seals every 2,000 hours or upon detection of > 10 drops/minute; use API Plan 53A dual mechanical seal configuration with barrier fluid system; maintain seal flush oil pressure 50 PSI above compressor discharge pressure

SECTION 4 - INVENTORY DATA

- **Current Stock Levels:** Stage 1 compressor (2 units), Stage 2 compressor (1 unit), Stage 3 compressor (1 unit); total spare capacity covers 100% redundancy for single-stage failure; bearing cartridge assemblies (4 units); mechanical seal kits (6 units); inlet filter elements (12 units)
- **Reorder Thresholds & Lead Times:** Stage 1 reorder at ≤ 1 unit remaining (8-10 week lead time); Stage 2 reorder at ≤ 0 units (10-12 week lead time); Stage 3 reorder at ≤ 0 units (12-14 week lead time); bearing cartridges reorder at ≤ 2 units (4-6 weeks); seal kits reorder at ≤ 3 units (3-4 weeks)
- **Primary Supplier & Backup:** Gulf Coast Industrial Supply (primary) | Backup: Midwest Turbomachinery Solutions, Kansas City MO (lead time 14-16 weeks); all suppliers maintain API 6A certification; purchase orders require 30-day advance notice for expedited delivery
- **Annual Budget Allocation & Spare Parts Cost:** Estimated annual spare parts budget \$185,000 USD; compressor stage replacement cost \$42,000-\$58,000 per unit; bearing cartridge \$8,500; mechanical seal kit \$3,200; filter elements \$450 per set; maintain minimum 18-month supply of critical consumables per company policy

Document Prepared By: Marcus J. Whitfield, Field Engineer

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Compliance Status: ✓ OSHA 1910 | ✓ API RP 54 | ✓ API 6A

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